

Answer the questions in the spaces provided. If you run out of room for an answer, continue on the back of the page.

1. Given the equation $x^n + y^n = z^n$ for x, y, z and n positive integers.
 - (a) For what values of n is the statement in the previous question true?

 - (b) For $n = 2$ there's a theorem with a special name. What's that name?

 - (c) What famous mathematician had an elegant proof for this theorem but there was not enough space in the margin to write it down?

 - i. Who actually proved the theorem?

 - ii. How long did actually take to solve this problem?

2. Prove that the real part of all non-trivial zeros of the function $\zeta(z)$ is $\frac{1}{2}$...